

ONTREX 100mg Powder for Concentrate for Solution for Infusion
ONTREX 500mg Powder for Concentrate for Solution for Infusion

1. NAME OF THE MEDICINAL PRODUCT

ONTREX 100mg Powder for Concentrate for Solution for Infusion
ONTREX 500mg Powder for Concentrate for Solution for Infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

ONTREX 100mg Powder for Concentrate for Solution for Infusion:

Each 100mg vial contains 100mg of pemetrexed (as pemetrexed disodium 2.5 Hydrate).

Each vial must be reconstituted with 4.2ml of 0.9% sodium chloride injection resulting in 25mg/ml solution. The appropriate volume of required dose is removed from the vial and further diluted to 100ml with sodium chloride injection.

ONTREX 500mg Powder for Concentrate for Solution for Infusion:

Each 500mg vial contains 500mg of pemetrexed (as pemetrexed disodium 2.5 Hydrate).

Each vial must be reconstituted with 20ml of 0.9% sodium chloride injection resulting in 25mg/ml solution. The appropriate volume of required dose is removed from the vial and further diluted to 100ml with sodium chloride injection.

3. PHARMACEUTICAL FORM

Lyophilised powder for concentrate for solution for infusion.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

Malignant Pleural Mesothelioma:

ONTREX in combination with cisplatin is indicated for the treatment of chemotherapy naive patients with unresectable malignant pleural mesothelioma.

Non-small Cell Lung Cancer:

ONTREX in combination with cisplatin is indicated for the first line treatment of patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) other than predominantly squamous cell histology.

ONTREX is indicated as monotherapy for the maintenance treatment of locally advanced or metastatic non-small cell lung cancer other than predominantly squamous cell histology in patients whose disease has not progressed immediately following platinum-based chemotherapy.

ONTREX is indicated as monotherapy for the second-line treatment of patients with locally advanced or metastatic non-small cell lung cancer other than predominantly squamous cell histology.

ONTREX in combination with pembrolizumab and platinum chemotherapy, is indicated for the first line treatment of patients with metastatic non-squamous NSCLC, with no EGFR or ALK genomic tumor aberrations.

4.2 Posology and Method of Administration

Posology

ONTREX must only be administered under the supervision of a physician qualified in the use of anti-cancer chemotherapy.

ONTREX must only be prepared according to the instructions provided in section 6.6.

ONTREX in combination with cisplatin

The recommended dose of ONTREX is 500mg/m² of body surface area (BSA) administered as an intravenous infusion over 10 minutes on the first day of each 21-day cycle. The recommended dose of cisplatin is 75mg/m² BSA infused over two hours approximately 30 minutes after completion of the pemetrexed infusion on the first day of each 21-day cycle.

Patients must receive adequate anti-emetic treatment and appropriate hydration prior to and/or after receiving cisplatin.

ONTREX as single agent

In patients treated for non-small cell lung cancer after prior chemotherapy, the recommended dose of ONTREX is 500mg/m² BSA administered as an intravenous infusion over 10 minutes on the first day of each 21-day cycle.

ONTREX in combination with pembrolizumab

The recommended dose of ONTREX when administered with pembrolizumab and platinum chemotherapy for the initial treatment of metastatic non-squamous NSCLC in patients with a creatinine clearance (calculated by Cockcroft-Gault equation) of 45 mL/min or greater is 500 mg/m² as an intravenous infusion over 10 minutes administered after pembrolizumab and prior to carboplatin or cisplatin on Day 1 of each 21-day cycle for 4 cycles. Following completion of platinum-based therapy, treatment with ONTREX with or without pembrolizumab is administered until disease progression or unacceptable toxicity.

Pre-medication Regimen

To reduce the incidence and severity of skin reactions, a corticosteroid should be given the day prior to, on the day of, and the day after pemetrexed administration. The corticosteroid should be equivalent to 4mg of dexamethasone administered orally twice a day (see section 4.4).

To reduce toxicity, patients treated with pemetrexed must also receive vitamin supplementation (see section 4.4). Patients must take oral folic acid or a multivitamin containing folic acid (350 to 1,000 micrograms) on a daily basis. At least five doses of folic acid must be taken during the seven days preceding the first dose of pemetrexed, and dosing must continue during the full course of therapy and for 21 days after the last dose of pemetrexed. Patients must also receive an intramuscular injection of vitamin B₁₂ (1,000 micrograms) in the week preceding the first dose of pemetrexed and once every three cycles thereafter. Subsequent vitamin B₁₂ injections may be given on the same day as pemetrexed.

Monitoring

Patients receiving pemetrexed should be monitored before each dose with a complete blood count, including a differential white cell count (WCC) and platelet count. Prior to each chemotherapy administration, blood chemistry tests should be collected to evaluate renal and hepatic function. Before the start of any cycle of chemotherapy, patients are required to have the following: absolute neutrophil count (ANC) should be $\geq 1,500$ cells/mm³ and platelets should be $\geq 100,000$ cells/mm³.

Creatinine clearance should be ≥ 45 ml/min.

The total bilirubin should be ≤ 1.5 times upper limit of normal. Alkaline phosphatase (AP), aspartate aminotransferase (AST or SGOT) and alanine aminotransferase (ALT or SGPT) should be ≤ 3 times upper limit of normal. Alkaline phosphatase, AST, and ALT ≤ 5 times upper limit of normal is acceptable if liver has tumour involvement.

Dose Adjustments

Dose adjustments at the start of a subsequent cycle should be based on nadir haematologic counts or maximum non-haematologic toxicity from the preceding cycle of therapy. Treatment may be delayed to allow sufficient time for recovery. Upon recovery, patients should be re-treated using the guidelines in Tables 1, 2, and 3, which are applicable for ONTREX used as a single agent or in combination with cisplatin.

Table 1. Dose Modification Table for ONTREX (as Single Agent or in Combination) and Cisplatin - Haematologic Toxicities	
Nadir ANC < 500/mm ³ and nadir platelets ≥ 50,000/mm ³	75% of previous dose (both ONTREX and cisplatin)
Nadir platelets < 50,000/mm ³ regardless of nadir ANC	75% of previous dose (both ONTREX and cisplatin)
Nadir platelets < 50,000/mm ³ with bleeding ^a , regardless of nadir ANC	50% of previous dose (both ONTREX and cisplatin)
^a These criteria meet the National Cancer Institute Common Toxicity Criteria (CTC v2.0; NCI 1998) definition of ≥ CTC Grade 2 bleeding.	

If patients develop non-haematologic toxicities ≥ Grade 3 (excluding neurotoxicity), ONTREX should be withheld until resolution to less than or equal to the patient's pre-therapy value. Treatment should be resumed according to the guidelines in Table 2.

Table 2. Dose Modification Table for ONTREX (as Single Agent or in Combination) and Cisplatin - Non-Haematologic Toxicities^{a, b}		
	Dose of ONTREX (mg/m²)	Dose for Cisplatin (mg/m²)
Any Grade 3 or 4 toxicities except mucositis	75% of previous dose	75% of previous dose
Any diarrhoea requiring hospitalisation (irrespective of grade) or Grade 3 or 4 diarrhoea	75% of previous dose	75% of previous dose
Grade 3 or 4 mucositis	50% of previous dose	100% of previous dose
^a National Cancer Institute Common Toxicity Criteria (CTC)		
^b Excluding neurotoxicity		

In the event of neurotoxicity, the recommended dose adjustment for ONTREX and cisplatin is documented in Table 3. Patients should discontinue therapy if Grade 3 or 4 neurotoxicity is observed.

Table 3. Dose Modification Table for ONTREX (as Single Agent or in Combination) and Cisplatin – Neurotoxicity		
CTC^a Grade	Dose of ONTREX (mg/m²)	Dose for Cisplatin (mg/m²)
0 - 1	100% of previous dose	100% of previous dose
2	100% of previous dose	50% of previous dose
^a National Cancer Institute Common Toxicity Criteria (CTC v2.0; NCI 1998)		

Treatment with ONTREX should be discontinued if a patient experiences any haematologic or non-haematologic Grade 3 or 4 toxicity after 2 dose reductions or immediately if Grade 3 or 4 neurotoxicity is observed.

Elderly

There has been no indication that patients 65 years of age or older are at increased risk of adverse events compared to patients younger than 65 years old. No dose reductions other than those recommended for all patients are necessary.

Paediatric population

ONTREX is not recommended for use in patients under 18 years of age, as safety and efficacy have not been established in this group of patients.

Patients with renal impairment (standard Cockcroft and Gault formula or Glomerular Filtration Rate measured Tc99m DPTA serum clearance method)

Pemetrexed is primarily eliminated unchanged by renal excretion. Patients with creatinine clearance of ≥ 45 ml/min required no dose adjustments other than those recommended for all patients. There are insufficient data on the use of pemetrexed in patients with creatinine clearance below 45 ml/min; therefore, the use of pemetrexed is not recommended (see section 4.4).

Patients with hepatic impairment

No relationships between AST (SGOT), ALT (SGPT), or total bilirubin and pemetrexed pharmacokinetics were identified. However, patients with hepatic impairment, such as bilirubin > 1.5 times the upper limit of normal and/or aminotransferase > 3.0 times the upper limit of normal (hepatic metastases absent) or > 5.0 times the upper limit of normal (hepatic metastases present), have not been specifically studied.

Route of Administration

Intravenous (IV)

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

Breast-feeding (see section 4.6).

Concomitant yellow fever vaccine (see section 4.5).

4.4 Special Warnings and Precautions for Use

Pemetrexed can suppress bone marrow function as manifested by neutropenia, thrombocytopenia, and anaemia (or pancytopenia) (see section 4.8). Myelosuppression is usually the dose-limiting toxicity. Patients should be monitored for myelosuppression during therapy and pemetrexed should not be given to patients until absolute neutrophil count (ANC) returns to $\geq 1,500$ cells/mm³ and platelet count returns to $\geq 100,000$ cells/mm³. Dose reductions for subsequent cycles are based on nadir ANC, platelet count, and maximum non-haematologic toxicity seen from the previous cycle (see section 4.2).

All patients treated with pemetrexed must be instructed to take folic acid and vitamin B₁₂ as a prophylactic measure to reduce treatment-related toxicity (see section 4.2).

Skin reactions have been reported in patients not pre-treated with a corticosteroid. Pre-treatment with dexamethasone (or equivalent) can reduce the incidence and severity of skin reactions (see section 4.2).

The use of pemetrexed in patients with creatinine clearance of <45 ml/min is not recommended (see section 4.2).

Patients with mild to moderate renal insufficiency (creatinine clearance from 45 to 79 ml/min) should avoid taking non-steroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen, and acetylsalicylic acid (>1.3 g daily) for 2 days before, on the day of, and 2 days following pemetrexed administration (see section 4.5).

In patients with mild to moderate renal insufficiency eligible for pemetrexed therapy, NSAIDs with long elimination half-lives should be interrupted for at least 5 days prior to, on the day of, and at least 2 days following pemetrexed administration (see section 4.5).

Serious renal events, including acute renal failure, have been reported with pemetrexed alone or in association with other chemotherapeutic agents. Many of the patients in whom these occurred had underlying risk factors for the development of renal events, including dehydration or pre-existing hypertension or diabetes. Nephrogenic diabetes insipidus and renal tubular necrosis were also reported in post marketing setting with pemetrexed alone or with other chemotherapeutic agents. Most of these events resolved after pemetrexed withdrawal. Patients should be regularly monitored for acute tubular necrosis, decreased renal function and signs and symptoms of nephrogenic diabetes insipidus (e.g. hypernatraemia).

The effect of third-space fluid, such as pleural effusion or ascites, on pemetrexed is not fully defined. Thus, drainage of third-space fluid collection prior to pemetrexed treatment should be considered, but may not be necessary.

Due to the gastrointestinal toxicity of pemetrexed given in combination with cisplatin, severe dehydration has been observed. Therefore, patients should receive adequate anti-emetic treatment and appropriate hydration prior to and/or after receiving treatment.

Serious cardiovascular events, including myocardial infarction and cerebrovascular events, have been uncommonly reported with pemetrexed, usually when given in combination with another cytotoxic agent. Most of the patients in whom these events have been observed had pre-existing cardiovascular risk factors (see section 4.8).

Immunodepressed status is common in cancer patients. As a result, concomitant use of live attenuated vaccines is not recommended (see section 4.5).

Pemetrexed can have genetically damaging effects. Sexually mature males are advised not to father a child during the treatment and up to 6 months thereafter. Contraceptive measures or abstinence are recommended. Owing to the possibility of pemetrexed treatment causing irreversible infertility, men are advised to seek counselling on sperm storage before starting treatment.

Women of childbearing potential must use effective contraception during treatment with pemetrexed and for 6 months following completion of treatment (see section 4.6).

Cases of radiation pneumonitis have been reported in patients treated with radiation either prior, during, or subsequent to their pemetrexed therapy. Particular attention should be paid to these patients, and caution exercised with use of other radiosensitising agents.

Cases of radiation recall have been reported in patients who received radiotherapy weeks or years previously.

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

Pemetrexed is mainly eliminated unchanged renally by tubular secretion and to a lesser extent by glomerular filtration. Concomitant administration of nephrotoxic drugs (e.g., aminoglycoside, loop diuretics, platinum compounds, cyclosporin) could potentially result in delayed clearance of pemetrexed. This combination should be used with caution. If necessary, creatinine clearance should be closely monitored.

Concomitant administration of substances that are also tubularly secreted (e.g., probenecid, penicillin) could potentially result in delayed clearance of pemetrexed. Caution should be made when these drugs are combined with pemetrexed. If necessary, creatinine clearance should be closely monitored.

In patients with normal renal function (creatinine clearance ≥ 80 ml/min), high doses of non-steroidal anti-inflammatory drugs (NSAIDs, such as ibuprofen > 1600 mg/day) and acetylsalicylic acid at higher doses (≥ 1.3 g daily) may decrease pemetrexed elimination and, consequently, increase the

occurrence of pemetrexed adverse events. Therefore, caution should be made when administering higher doses of NSAIDs or acetylsalicylic acid, concurrently with pemetrexed to patients with normal function (creatinine clearance $\geq 80\text{ml/min}$).

In patients with mild to moderate renal insufficiency (creatinine clearance from 45 to 79ml/min), the concomitant administration of pemetrexed with NSAIDs (e.g., ibuprofen) or acetylsalicylic acid at higher doses should be avoided for 2 days before, on the day of, and 2 days following pemetrexed administration (see section 4.4).

In the absence of data regarding potential interaction with NSAIDs having longer half-lives such as piroxicam or rofecoxib, the concomitant administration with pemetrexed in patients with mild to moderate renal insufficiency should be interrupted for at least 5 days prior to, on the day of, and at least 2 days following pemetrexed administration (see section 4.4). If concomitant administration of NSAIDs is necessary, patients should be monitored closely for toxicity, especially myelosuppression and gastrointestinal toxicity.

Pemetrexed undergoes limited hepatic metabolism. Pemetrexed would not be predicted to cause clinically significant inhibition of the metabolic clearance of drugs metabolised by CYP3A, CYP2D6, CYP2C9, and CYP1A2.

Interactions Common to all Cytotoxics

Due to the increased thrombotic risk in patients with cancer, the use of anticoagulation treatment is frequent. The high intra-individual variability of the coagulation status during diseases and the possibility of interaction between oral anticoagulants and anti-cancer chemotherapy require increased frequency of INR (International Normalised Ratio) monitoring, if it is decided to treat the patient with oral anticoagulants.

Concomitant Use Contraindicated:

Yellow fever vaccine: Risk of fatal generalised vaccinale disease (see section 4.3).

Concomitant Use Not Recommended:

Live attenuated vaccines (except yellow fever, for which concomitant use is contraindicated): Risk of systemic, possibly fatal, disease. The risk is increased in subjects who are already immunosuppressed by their underlying disease. Use an inactivated vaccine where it exists (poliomyelitis) (see section 4.4).

4.6 Fertility, Pregnancy and Lactation

Women of childbearing potential/Contraception in males and females

Pemetrexed can have genetically damaging effects. Women of childbearing potential must use effective contraception during treatment with pemetrexed and for 6 months following completion of treatment. Sexually mature males are advised to use effective contraceptive measures and not to father a child during the treatment and up to 3 months thereafter.

Pregnancy

There are no data from the use of pemetrexed in pregnant women; but pemetrexed, like other anti-metabolites, is suspected to cause serious birth defects when administered during pregnancy. Pemetrexed should not be used during pregnancy unless clearly necessary, after a careful consideration of the needs of the mother and the risk for the foetus (see section 4.4).

Breast-feeding

It is not known whether pemetrexed is excreted in human milk, and adverse reactions on the suckling child cannot be excluded. Breast-feeding must be discontinued during pemetrexed therapy (see section 4.3).

Fertility

Owing to the possibility of pemetrexed treatment causing irreversible infertility, men are advised to seek counselling on sperm storage before starting treatment.

4.7 Effects on Ability to Drive and Use Machines

No studies on the effects on the ability to drive and use machines have been performed. However, it has been reported that pemetrexed may cause fatigue. Therefore, patients should be cautioned against driving or operating machines if this event occurs.

4.8 Undesirable Effects

Summary of the safety profile

The most commonly reported undesirable effects related to pemetrexed, whether used as monotherapy or in combination, are bone marrow suppression manifested as anaemia, neutropenia, leukopenia, thrombocytopenia; and gastrointestinal toxicities, manifested as anorexia, nausea, vomiting, diarrhoea, constipation, pharyngitis, mucositis, and stomatitis. Other undesirable effects include renal toxicities, increased aminotransferases, alopecia, fatigue, dehydration, rash, infection/sepsis and neuropathy. Rarely seen events include Stevens-Johnson syndrome and Toxic epidermal necrolysis.

Tabulated list of adverse reactions

The table 4 lists the adverse drug events regardless of causality associated with pemetrexed used either as a monotherapy treatment or in combination with cisplatin from the studies (JMCH, JMEI, JMBD, JMEN and PARAMOUNT) and from the post marketing period.

ADRs are listed by MedDRA body system organ class. The following convention has been used for classification of frequency: very common; common; uncommon; rare; very rare and not known (cannot be estimated from the available data).

Table 4. Frequencies of all grades adverse drug events regardless of causality from the studies: JMEI, JMDB, JMCH, JMEN and PARAMOUNT and from post-marketing period.

System Organ Class (MedDRA)	Very common	Common	Uncommon	Rare	Very rare	Not known
Infections and infestations	Infection ^a Pharyngitis	Sepsis ^b			Dermo-hypodermatitis	
Blood and lymphatic system disorders	Neutropenia Leukopenia Haemoglobin decreased	Febrile neutropenia Platelet count decreased	Pancytopenia	Autoimmune haemolytic anaemia		
Immune system disorders		Hypersensitivity		Anaphylactic shock		
Metabolism and nutrition disorders		Dehydration				
Nervous system disorders		Taste disorder Peripheral motor neuropathy Peripheral sensory neuropathy Dizziness	Cerebrovascular accident Ischaemic stroke Haemorrhage intracranial			
Eye disorders		Conjunctivitis Dry eye				

		Lacrimation increased Keratoconjunctivitis sicca Eyelid oedema Ocular surface disease				
Cardiac disorders		Cardiac failure Arrhythmia	Angina Myocardial infarction Coronary artery disease Arrhythmia supraventricular			
Vascular disorders			Peripheral ischaemia ^c			
Respiratory, thoracic and mediastinal disorders			Pulmonary embolism Interstitial pneumonitis ^{bd}			
Gastrointestinal disorders	Stomatitis Anorexia Vomiting Diarrhoea Nausea	Dyspepsia Constipation Abdominal pain	Rectal haemorrhage Gastrointestinal haemorrhage Intestinal perforation Oesophagitis Colitis ^e			
Hepatobiliary disorders		Alanine aminotransferase increased Aspartate aminotransferase increased		Hepatitis		
Skin and subcutaneous tissue disorders	Rash Skin exfoliation	Hyperpigmentation Pruritus Erythema multiforme Alopecia Urticaria		Erythema	Stevens-Johnson syndrome ^b Toxic epidermal necrolysis ^b Pemphigoid Dermatitis bullous Acquired epidermolysis bullosa Erythematous oedema ^f Pseudocellulitis Dermatitis Eczema Prurigo	
Renal and urinary	Creatinine clearance	Renal failure Glomerular				Nephrogenic diabetes

disorders	decreased Blood creatinine increased ^e	filtration rate decreased				insipidus Renal tubular necrosis
General disorders and administration site conditions	Fatigue	Pyrexia Pain Oedema Chest pain Mucosal inflammation				
Investigations		Gamma-glutamyltransferase increased				
Injury, poisoning and procedural complications			Radiation oesophagitis Radiation pneumonitis	Recall phenomenon		

^a with and without neutropenia

^b in some cases fatal

^c sometimes leading to extremity necrosis

^d with respiratory insufficiency

^e seen only in combination with cisplatin

^f mainly of the lower limbs

4.9 Overdose

Reported symptoms of overdose include neutropenia, anaemia, thrombocytopenia, mucositis, sensory polyneuropathy, and rash. Anticipated complications of overdose include bone marrow suppression as manifested by neutropenia, thrombocytopenia, and anaemia. In addition, infection with or without fever, diarrhoea, and/or mucositis may be seen. In the event of suspected overdose, patients should be monitored with blood counts and should receive supportive therapy as necessary. The use of calcium folinate/folinic acid in the management of pemetrexed overdose should be considered.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Pharmacotherapeutic group: Folic acid analogues. ATC code: L01BA04.

Pemetrexed is a multi-targeted anti-cancer antifolate agent that exerts its action by disrupting crucial folate-dependent metabolic processes essential for cell replication.

In vitro studies have shown that pemetrexed behaves as a multi-targeted antifolate by inhibiting thymidylate synthase (TS), dihydrofolate reductase (DHFR), and glycinamide ribonucleotide formyltransferase (GARFT), which are key folate-dependent enzymes for the *de novo* biosynthesis of thymidine and purine nucleotides. Pemetrexed is transported into cells by both the reduced folate carrier and membrane folate binding protein transport systems. Once in the cell, pemetrexed is rapidly and efficiently converted to polyglutamate forms by the enzyme folylpolyglutamate synthetase. The polyglutamate forms are retained in cells and are even more potent inhibitors of TS and GARFT. Polyglutamation is a time- and concentration-dependent process that occurs in tumour cells and, to a lesser extent, in normal tissues.

Polyglutamated metabolites have an increased intracellular half-life resulting in prolonged drug action

in malignant cells.

5.2 Pharmacokinetic Properties

The pharmacokinetic properties of pemetrexed following single-agent administration have been evaluated in cancer patients with a variety of solid tumours at doses ranging from 0.2 to 838mg/m² infused over a 10-minute period.

Pemetrexed has a steady-state volume of distribution of 9 l/m². *In vitro* studies indicate that pemetrexed is approximately 81% bound to plasma proteins. Binding was not notably affected by varying degrees of renal impairment. Pemetrexed undergoes limited hepatic metabolism. Pemetrexed is primarily eliminated in the urine, with 70% to 90% of the administered dose being recovered unchanged in urine within the first 24 hours following administration. *In vitro* studies indicate that pemetrexed is actively secreted by OAT3 (organic anion transporter).

Pemetrexed total systemic clearance is 91.8ml/min and the elimination half-life from plasma is 3.5 hours in patients with normal renal function (creatinine clearance of 90 ml/min). Between-patient variability in clearance is moderate at 19.3%. Pemetrexed total systemic exposure (AUC) and maximum plasma concentration increase proportionally with dose. The pharmacokinetics of pemetrexed are consistent over multiple treatment cycles.

The pharmacokinetic properties of pemetrexed are not influenced by concurrently administered cisplatin. Oral folic acid and intramuscular vitamin B₁₂ supplementation do not affect the pharmacokinetics of pemetrexed.

6 PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Mannitol

Hydrochloric acid

Sodium hydroxide

6.2 Incompatibilities

Pemetrexed is physically incompatible with diluents containing calcium, including lactated Ringer's injection and Ringer's injection. In the absence of other compatibility studies this medicinal product must not be mixed with other medicinal products.

6.3 Shelf Life

Unopened vial 100mg & 500mg: 3 years

Reconstitution and infusion solutions

Chemical and physical stability of reconstitution and infusion solutions of ONTREX 100mg & 500mg Powder for Concentrate for Solution for Infusion were demonstrated for up to 24 hours following initial reconstitution, when stored refrigerated, 2° to 8°C (36° to 46°F). When prepared as directed, reconstitution and infusion solutions of ONTREX 100mg & 500mg Powder for Concentrate for Solution for Infusion contain no antimicrobial preservatives. Discard unused portion.

ONTREX 100mg & 500mg Powder for Concentrate for Solution for Infusion is not light sensitive.

6.4 Special Precautions for Storage

Store below 30°C for unopened vial.

Reconstituted and infusion solution:

- When prepared as directed, reconstituted and infusion solutions contain no antimicrobial preservatives. Discard unused portion.
- Chemical and physical in-use stability of reconstituted and infusion solutions of ONTREX 100mg & 500mg Powder for Concentrate for Solution for Infusion were demonstrated for 24 hours at refrigerated temperature.
- From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would not be longer than 24 hours at 2°C to 8°C.

6.5 Nature and Content of Container

Pack of 1 vial.

ONTREX 100mg Powder for Concentrate for Solution for Infusion:

Type I, Clear Borosilicate glass vials containing 120.8mg pemetrexed disodium hemipentahydrate (equivalent to 100 mg pemetrexed), and sealed with a 20-D2 lyophilization Bromobutyl Rubber stopper, and secured with aluminum flip-off seal with blue plastic over-seal.

ONTREX 500mg Powder for Concentrate for Solution for Infusion:

Type I, Clear Borosilicate glass vials containing 604.1mg pemetrexed disodium hemipentahydrate (equivalent to 500 mg pemetrexed), and sealed with a 20-D2 lyophilization Bromobutyl Rubber stopper, and secured with aluminum flip-off seal with blue plastic over-seal.

6.6 Special precautions for disposal and other handling

Pemetrexed solutions are for single use only.

Preparation and administration precautions

As with other potentially toxic anticancer agents, care should be exercised in the handling and preparation of pemetrexed infusion solutions. The use of gloves is recommended. If a pemetrexed solution contacts the skin, wash the skin immediately and thoroughly with soap and water. If pemetrexed solutions contact the mucous membranes, flush thoroughly with water. Pemetrexed is not a vesicant. There is not a specific antidote for extravasation of pemetrexed. There have been few reported cases of pemetrexed extravasation, which were not assessed as serious by the investigator. Extravasation should be managed by local standard practice as with other non-vesicants.

1. Use aseptic technique during the reconstitution and further dilution of ONTREX for intravenous infusion administration.
2. Calculate the dose of ONTREX and determine the number of vials needed. For ONTREX 100mg Powder for Concentrate for Solution for Infusion, vials contain 100 mg of pemetrexed. For ONTREX 500mg Powder for Concentrate for Solution for Infusion, vials contain 500 mg of pemetrexed. The vials contain an excess of pemetrexed to facilitate delivery of label amount.
3. **ONTREX 100mg Powder for Concentrate for Solution for Infusion:** Reconstitute each 100-mg vial with 4.2 ml of 0.9% Sodium Chloride Injection (preservative free).
ONTREX 500mg Powder for Concentrate for Solution for Infusion: Reconstitute each 500-mg vial with 20 ml of 0.9% Sodium Chloride Injection (preservative free).
Reconstitution of vial gives a solution containing 25 mg/ml pemetrexed. Gently swirl each vial until the powder is completely dissolved. The resulting solution is clear and ranges in color from colorless to yellow or green-yellow without adversely affecting product quality. The pH of the

reconstituted ONTREX solution is between 6.6 and 7.8. FURTHER DILUTION IS REQUIRED.

4. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit. If particulate matter is observed, do not administer.
5. An appropriate quantity of the reconstituted ONTREX solution must be further diluted into a solution of 0.9% Sodium Chloride Injection (preservative free), so that the total volume of solution is 100 ml. ONTREX is administered as an intravenous infusion over 10 minutes.
6. Chemical and physical stability of reconstituted and infusion solutions of ONTREX were demonstrated for up to 24 hours following initial reconstitution, when stored refrigerated. When prepared as directed, reconstitution and infusion solutions of ONTREX contain no antimicrobial preservatives. Discard any unused portion.

Reconstitution and further dilution prior to intravenous infusion is only recommended with 0.9% Sodium Chloride Injection (preservative free). ONTREX is physically incompatible with diluents containing calcium, including Lactated Ringer's Injection, USP and Ringer's Injection, USP and therefore these should not be used. Coadministration of ONTREX with other drugs and diluents has not been studied, and therefore is not recommended. ONTREX is compatible with standard polyvinyl chloride (PVC) administration sets and non-PVC multilayer film intravenous bags.

**CAUTION: CYTOTOXIC AGENT
CONTROLLED MEDICINE**

**KEEP MEDICINE OUT OF REACH OF CHILDREN
JAUHI UBAT DARIPADA KANAK-KANAK**

7 MANUFACTURER

Manufactured by:

Jiangsu Hansoh Pharmaceutical Group Co., Ltd.
No.5, Dong Jin Road,
Economic & Technical Development Zone,
Lianyungang, Jiangsu 222069, China.

Product Registration Holder:

Novugen Oncology Sdn. Bhd.
No. 3, Jalan Jururancang U1/21,
Hicom Glenmarie Industrial Park,
40150 Shah Alam, Selangor, Malaysia

8 DATE OF REVISION

19/07/2024

Package insert is drafted in Times New Roman with font size of 11, subject to change based on any other legible font and font size.
